

4(a)	quartz crystal	B1
	alternating p.d. across crystal causes it to vibrate	B1
	resonance occurs when frequency of p.d. matches natural frequency of crystal	B1
	natural frequency of crystal is in ultrasound range	B1
4(b)	$I = I_0 e^{-\mu x}$	C1
	$I / I_0 = e^{-1.2 \times 3.5}$ $= 0.015$	C1
	ratio / dB = $-10 \lg (1 / 0.015)$ or $10 \lg (0.015)$	C1
	$= -18 \text{ dB}$	A1